Reinforcement Learning

Lecture 1: Introduction and Overview

SJTU

Motivating Example: Robot Grasping

- **Goal:** Build a system for robots to pick up diverse objects
- Robot sees camera images $\rightarrow$ outputs grasp coordinates
- Many edge cases:
  - Rigid vs. deformable objects
  - Center of mass, friction, shape variety
  - Novel objects never seen before
- Machine learning is attractive, but...
- Supervised learning requires **labeled datasets of optimal grasps**
- Where do we get ground-truth labels?

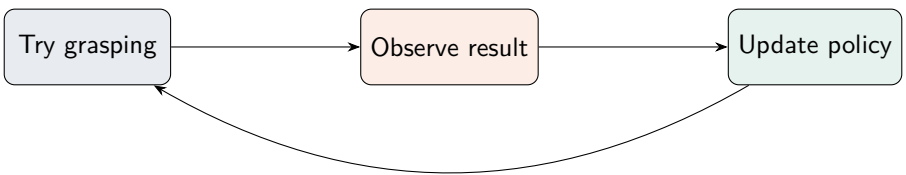
Why Not Just Supervised Learning?

- Supervised learning needs labeled $(\mathbf{x}, \mathbf{y})$ pairs with **ground truth**
- For grasping: even humans can't easily specify the optimal grasp position
- SL assumes:
  - Data is **i.i.d.** (independently and identically distributed)
  - Ground-truth outputs are **known**
- Neither assumption holds for many real-world decision-making problems

We need a framework where the agent learns from **interaction**, not just from labels.

The RL Approach

- Let the robot **collect its own data** through trial and error
- Label with **success / failure** (reward), not optimal actions
- The RL algorithm learns to **maximize reward**
- Can potentially **exceed average human behavior**



Recent AI Advances

- **Image generation:** Diffusion models (DALL-E, Stable Diffusion)
- **Large language models:** Conversation, reasoning, coding
- **Protein structure prediction:** AlphaFold
- **Common theme:** All based on density estimation

Task	**Learning objective**
Image generation	$p_\theta(\mathbf{x})$
Language modeling	$p_\theta(\mathbf{y} \mid \mathbf{x})$
Protein folding	$p_\theta(\text{structure} \mid \text{sequence})$

How Data-Driven AI Works

- Learns the **distribution of human-generated data**
- Images: what people put online
- Text: what people write
- Great for generating **human-like** content

Data-driven AI doesn't try to do **better** than the data.

- It mimics what it has seen — it does not *optimize* beyond the training distribution

Two Roots of Modern RL

Psychology

- B.F. Skinner: operant conditioning
- Animals adapt behavior via rewards and punishments
- Agent learns from environmental feedback

Control & Optimization

- Karl Sims (1994): virtual creatures
- Emergent locomotion and competition
- Optimization of complex behaviors

Modern Deep RL = Large-scale optimization + Classical RL algorithms

RL Can Discover New Solutions

- **AlphaGo "Move 37"**: Surprised expert Go commentators
  - A move no human professional had considered
  - Turned out to be brilliant
- **Data-driven AI** is impressive because it *looks like* what a human might produce
- **RL** is impressive because *no human had thought of it*

RL enables **emergent behavior** — solutions that go beyond human intuition.

What is Reinforcement Learning?

- A **mathematical formalism** for learning-based decision making
- An approach for learning decision making and control **from experience**

Important Distinction

Don't confuse the RL *problem* with the RL *solution*.

- The **problem**: sequential decision making under uncertainty
- The **solution**: a specific algorithm (Q-learning, policy gradient, etc.)

How is RL Different from Supervised Learning?

Standard (Supervised) ML

- Given $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}$
- Learn to predict $\mathbf{y}$ from $\mathbf{x}$:

$$f(\mathbf{x}) \approx \mathbf{y}$$

- Assumes:
  - Data is **i.i.d.**
  - Ground truth outputs are **known**

Reinforcement Learning

- Data is **NOT i.i.d.**: previous outputs influence future inputs
- Ground truth answer is **not known**
  - Only know if we *succeeded* or *failed*
- More generally: we know the **reward**

RL Formalism: SL vs RL

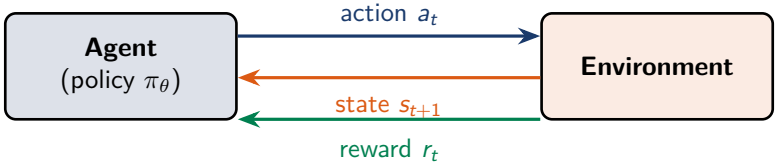
Supervised Learning

- **Input:** $\mathbf{x}$
- **Output:** $\mathbf{y}$
- **Data:** $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}$
  (someone gives this to you)
- **Goal:** $f_\theta(\mathbf{x}_i) \approx \mathbf{y}_i$

Reinforcement Learning

- **Input:** $\mathbf{s}_t$ at each time step
- **Output:** $\mathbf{a}_t$ at each time step
  (pick your own actions)
- **Data:** $(\mathbf{s}_1, \mathbf{a}_1, r_1, \ldots, \mathbf{s}_T, \mathbf{a}_T, r_T)$
- **Goal:** Learn $\pi_\theta : \mathbf{s}_t \rightarrow \mathbf{a}_t$ to maximize $\sum_t r_t$

The Agent–Environment Loop



Key elements:

- Decisions (actions) chosen by the agent
- Consequences & observations (states) returned by the environment
- Rewards: scalar feedback signal

RL Examples

Domain	Actions	Observations	Rewards
Dog training	Muscle contractions	Sight, smell, touch	Food treats
Robot control	Motor torques / velocities	Camera images, joint angles	Task success signal
Inventory mgmt	What / how much to purchase	Current inventory levels	Profit

The Credit Assignment Problem

- Reward may be **delayed**: which action actually caused the outcome?
- Example: bad grade at end of semester
  - Not from checking grades (last action)
  - From earlier decisions (not studying, skipping lectures)
- Must learn from data **without optimal output labels**
- Rewards may arrive **long after** the relevant actions

Goal: Obtain behavior **better** than what was previously observed.

Key Differences Summary

Aspect	**Supervised Learning**	**Reinforcement Learning**
Data	i.i.d., given by someone	Sequential, self-collected
Labels	Ground truth outputs	Reward signal (possibly delayed)
Goal	Match labels exactly	Maximize cumulative reward
Feedback	Immediate, exact	Delayed, scalar
Distribution	Fixed	Changes with agent's policy

Terminology Recap

State s_t Observation of the environment at time t

Action a_t Decision made by the agent at time t

Reward r_t Scalar feedback signal after taking a_t in s_t

Policy $\pi_\theta(a \mid s)$ Mapping from states to actions (the agent's "brain")

Return Cumulative reward: $G_t = \sum_{k=0}^{T} r_{t+k}$

The goal of RL: Find π_θ that maximizes $\mathbb{E}\left[\sum_{t=0}^{T} r_t\right]$

Complex Physical Tasks

- **Dexterous manipulation:** Learning to use tools (hammer + nail)
  - Rajeswaran et al., 2018
- Learning control policies for physically complex tasks
- **Key insight:** Easy to *define* the reward (nail is in), hard to manually specify *how* to achieve it
- RL discovers the motor strategies automatically

Agile Robot Locomotion

- Quadruped robots learning agile skills:
  - Jumping over obstacles
  - Standing on hind legs
  - Navigating uneven terrain
- **Sim-to-real transfer** (Smith et al., 2022)
  - Train in simulation, deploy on real hardware
- **End-to-end locomotion & navigation** (Rudin et al., 2022)
  - From raw sensor input to motor commands

Unexpected Solutions: Atari

- **DQN** learns to play Atari games from raw pixels (Mnih et al., 2015)
- In *Breakout*: discovers a clever strategy
  - Dig a tunnel through the bricks
  - Bounce the ball **above** the remaining bricks
  - Ball bounces repeatedly for maximum points
- This strategy was **not programmed**
- **Emergent** from pure reward maximization

RL finds solutions humans might not think of.

RL at Scale in the Real World

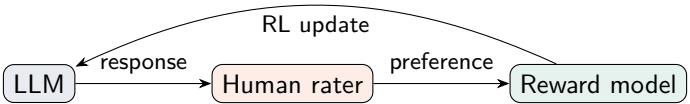
- **Waste sorting in office buildings** (Herzog et al., 2023)
  - Mobile manipulators sorting recycling
  - Vision-based manipulation skills
- Practicing in both **lab environments** and **real deployment**
- Challenges:
  - Safety during exploration
  - Diverse object types
  - Real-time decision making
- Demonstrates RL is moving beyond the lab

Traffic Control with RL

- Cathy Wu (MIT): RL for traffic flow optimization
- **Circular road experiment:**
  - Human driving models create spontaneous traffic jams
  - One RL-controlled vehicle slows slightly
  - $\Rightarrow$ Eliminates jams for **ALL** cars
- **Intersections:**
  - RL optimizes traffic light timing
  - Reduces wait times and congestion
- Small interventions $\rightarrow$ large systemic improvements

RL for Language Models (RLHF)

- Large language models trained on internet data
- Problem: most likely completion $\neq$ ideal response
- **RLHF** (RL from Human Feedback):
  - Train models based on human scoring / preferences
  - Shift from "most probable" to "most helpful"
- Key component of ChatGPT, Claude, and other assistants



RL for Image Generation

- **DDPO:** Training Diffusion Models with RL (Black et al., 2023)
- Example prompt: “a dolphin riding a bike”
- Pipeline:
  1. Diffusion model generates image
  2. Vision-language model (e.g., LLaVA) describes it
  3. Similarity to original prompt = **reward**
  4. RL optimizes the diffusion process
- Iterative improvement through RL optimization
- Images become increasingly aligned with text prompts

RL for Chip Design

- RL agent places chip macros **one at a time** on a chip canvas
- **State:** current placement of components
- **Action:** where to place the next macro
- **Reward:**
  - HPWL (half-perimeter wire length)
  - Congestion metrics
  - Timing constraints
- Fine-tuning a pre-trained policy converges **faster** than training from scratch
- Transfer learning across chip designs

Beyond Games and Robots

- **Scientific discovery**
  - Protein design, materials science, drug discovery
- **Resource allocation & scheduling**
  - Data center cooling, supply chain optimization
- **Personalized education & healthcare**
  - Adaptive tutoring, treatment planning
- **Finance**
  - Portfolio management, algorithmic trading

Any **sequential decision problem** with definable rewards is a candidate for RL.

Summary of RL Applications

Domain	**Actions**	**Observations**	**Rewards**
Robotics	Motor torques	Camera, sensors	Task success
Games	Game moves	Game state, pixels	Score / win
Traffic	Signal timing / speed	Traffic flow data	Throughput
NLP (RLHF)	Token generation	Context / prompt	Human preference
Image Gen	Denoising steps	Latent / pixel space	Prompt alignment
Chip Design	Macro placement	Current layout	Wire length

Data-Driven AI vs Reinforcement Learning

Data-Driven AI

All about using **data**

+ Learns about the real world from data
– Doesn't try to do **better** than the data

Reinforcement Learning

All about **optimization**

+ Optimizes a goal with emergent behavior
– Need to figure out how to use at scale!

Data without optimization doesn't allow us to solve new problems in new ways.

The Bitter Lesson (Richard Sutton, 2019)

"We have to learn the bitter lesson that building in how we think we think does not work in the long run."

- Two methods that **scale arbitrarily** with compute:
  1. **Learning**
  2. **Search**
- NOT "learning and GPUs" — but "learning and search"
- Hand-crafted features and rules always lose to scalable methods in the long run

Learning and Search

Learning

- Use **data** to extract **patterns**
- Allows us to **understand** the world
- Build internal models and representations

Search

- Use **computation** to extract **inferences**
- Some optimization process using iterative computation
- Leverages understanding for **emergence**



A Bit of Philosophy

- Why do we need machine learning anyway?

> "We need ML for one reason only — to produce **adaptable and complex decisions**."

- Daniel Wolpert (neuroscientist):
  "We have a brain for one reason only — to produce adaptable and complex movements."
- All ML problems are **decision problems** in disguise
  - Classification: decide which label
  - Generation: decide which token / pixel
  - Control: decide which action

How Do We Build Intelligent Machines?

- Where do you start? Study the brain?
- Option A: An algorithm for each "module" (vision, language, planning, ...)
- Option B: A **single flexible algorithm** that can do everything

Evidence for a single algorithm:

- Brain regions can **adapt**: visual cortex can process auditory input in blind individuals
- BrainPort: "seeing" with your tongue via electrical stimulation
- Suggests: the brain uses a **general-purpose** learning algorithm

Learning as the Basis of Intelligence

- Some things we can all do (walking, grasping)
- Some things we can only **learn** (driving, chess, programming)
- We can learn a **huge variety** of difficult things
- Therefore: our learning mechanisms are powerful enough for **everything we associate with intelligence**

Alan Turing: "Instead of trying to produce a program to simulate the adult mind, why not rather try to produce one which simulates the child's?"

What Must That Single Algorithm Do?

1. **Interpret rich sensory inputs**
   - Vision, language, touch, proprioception
2. **Choose complex actions**
   - Motor control, language generation, strategic planning

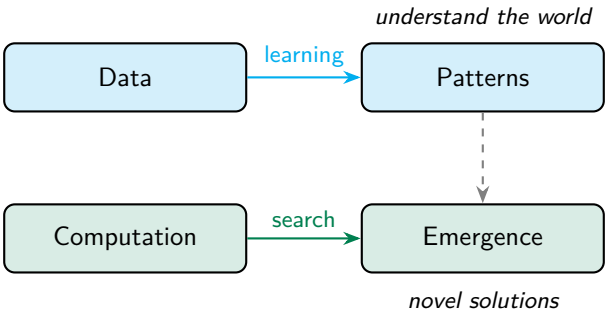
This is exactly what RL does:

observation $\xrightarrow{\pi_\theta}$ action

RL provides the most general framework for
perception $\rightarrow$ **decision** $\rightarrow$ **action** $\rightarrow$ **learning**

Why Deep Reinforcement Learning?

- **Deep** = scalable learning from large, complex datasets
- **Reinforcement Learning** = optimization through interaction
- **Together:** large-scale optimization with learned representations



Deep RL = Learning + Search

Summary and Key Takeaways

1. RL is a **mathematical formalism** for sequential decision making
2. Key differences from SL:
   - Non-i.i.d. data, delayed rewards, self-collected data
3. RL enables **emergent behavior** that surpasses human design
4. **Deep RL** = Learning (data $\rightarrow$ patterns) + Search (computation $\rightarrow$ emergence)
5. Applications span **robotics, games, NLP, image generation, chip design**, and more
6. The combination of data and optimization is the path to general intelligence

Next lecture: Markov Decision Processes and the RL objective.